WP N°	4			Lead beneficiary		WU	
WP title	Improvement of predictive breeding models in monogastrics						
Part. number	1	8	16	**21**			
Part. short name	INRAE	IRTA	UHOH	**WU**			
PM per Part.	29.1	11	6.5	**52**			
Start Month	M6			End Month		M60	

Objectives:
WP4 objectives are to develop and validate new models using epigenetic profiles and other functional information (e.g. gene expression) to:
- Dissect phenotypic variance into its heritable, genetic and non-genetic components
- Improve the accuracy of genomic prediction models
- Unravel trade-offs between ELP traits using multitrait and structural equation models

Description of work and role of participants
WP leader: WU (Mario Calus) / WP deputy leader: INRAE (Catherine Larzul)

Background
Genomic selection has been successfully used for more than a decade in many breeding programs around the world, because it enables to actually predict genetic merit early in life, allowing both fast and accurate genetic selection. More recently, several developments in the area of precision livestock farming have increased the desire to not only accurately predict genetic merit, but also phenotypes, which would allow farmers to provide to animals the environment that suits the best their (genetic) potential. In traditional genetic models, all factors other than genetics are included as nuisance factors to 'purify' estimates of genetic merit. Epigenetic information provided in GEroNIMO will be integrated in genomic evaluation models to evaluate whether or not this improves accuracy of predicted genomic breeding value for ELP traits. This will be achieved directly by exploiting epigenetic measures on studied individuals. Moreover, the part played by the inherited epigenetic variability in shaping phenotypes over generations is still unknown. Indeed, non-genetic sources of inheritance are currently not taken into account in livestock selection strategies. Nonetheless, if, as suggested in evolutionary studies, the non-genetic inheritance may be important for the transmission of phenotypes, then not considering non-genetic inheritance may result in suboptimal genetic progress and production in livestock populations.

Approach
WP4 aims at developing and validating innovative methods to improve selection strategies integrating environmental and non-genetic inherited factors, especially epigenetic profiles, for efficient livestock production (ELP) related traits in chicken and pig. This comprises models that enable to: i) dissect phenotypic variance in its underlying components, including epigenetic variance independent of genetic/genomic variance, ii) predict epigenetic profiles from (SNP) genotype data, iii) predict phenotypes and breeding values considering epigenetic profiles, next to other effects, such as genetic, permanent and common environment, iv) combine genetic and non-genetic sources of inheritance (with focus on epigenetic inheritance) to predict the global transmissible potential of animals, v) in addition include gene expression and epigenetic data (e.g. eQTL and QTM derived in WP1), vi) take into account trade-offs between ELP traits related to energy allocation, health and welfare in chickens and between fertility and production in pigs. Developed models will be applied to data generated in WP 1&2, including ELP phenotypes, epigenetic profiles and genomic information (pig, quail and laying hen), and gene expression data (pig and laying hen). In addition, outputs from WP4 will be used to organise an **EU level workshop** (WP6).

Work breakdown
Task 4.1 Jointly estimate heritable, genetic and non-genetic, components of phenotypic variation of complex traits considering epigenetic profiles (M6-M60**)**
Task leader: WU / Involved Partners: INRAE, IRTA
The aims of this task are to investigate the extent to which DNA methylation and other genetic and non-genetic factors contribute to (heritable) trait phenotypic variation. Several models will be developed to dissect the phenotypic variance of traits. To this end, epigenetic relationship matrices will be derived. One class of models will estimate variance components related to structural genomic information (i.e. SNPs) versus methylation patterns. Another will estimate inherited (genetic + non-genetic) phenotypic variance. Finally, multitrait models will be considered to evaluate the impact of methylation patterns on phenotypic and genetic correlations between traits, to investigate to what extent these contribute to trade-offs. To further elucidate trade-offs, causal relationship between ELP traits will be estimated using structural equation models. The various models developed will be

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applied to data produced on: i) sperm and meat quality traits in pig (Tasks 1.1 and 1.3); ii) productive longevity in laying hens (Task 1.2); iii) over several generations in quail (Task 2.4).
Links with other WPs: WP1, WP2, WP4, WP5, WP6

Task 4.2 Epigenetic and genomic prediction of phenotypes and breeding values of complex traits (M6-M60)
Task leader: IRTA / Involved Partners: INRAE, WU, UHOH
In this task parametric and non-parametric models that include epigenetic information for prediction of phenotypes and breeding values (genetic or global transmitted potential) of individuals for complex traits involved in ELP will be implemented and validated. This includes the models developed in task 4.1 and machine learning algorithms implemented in single or multiple-trait/multi-label models. The GBLUP will be used as a benchmark. The goal is to find the model leading to the highest prediction accuracy and stability of prediction quality across different validation sets. We will also evaluate 'preferential stability' (i.e. the agreement of prediction models produced by an algorithm when trained on different training sets), which is important to understand the biological processes involved. We will use records on ELP traits in pigs, quail and laying hens, epigenetic information for pigs and laying hens and recorded over several generations in quail (WP1/WP2 Task 2.4).
Links with other WPs: WP1, WP2, WP4, WP6

Deliverables
D4.1: Empirical validation of models considering epigenetic profiles when dissecting phenotypic variance in its underlying sources. M48: This deliverable will describe empirical validation of all the models developed in Task 4.1, using data generated in WP1 and WP2. This includes demonstration of how the models work, as well as reporting the proportions of phenotypic variance explained by each source. Deliverable leader: WU / Dissem. level: PU / Report.

D4.2: **Models to define the causal relationships between main ELP traits**. M52: A report describing the relationships between ELP traits and identifying the factors, including epigenetic, that influence the trade-offs between traits, using data from WP1 and WP2. Deliverable leader: INRAE / Dissem. level: PU - Report

D4.3: **Models to predict phenotypes and breeding values based on genomic and epigenetic profiles** M60. A report with results regarding phenotype prediction quality of the different models evaluated will be delivered. Deliverable leader: IRTA / Dissem. level: PU / Report.

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